

Technical Report — From Batch Leakage to an Interpretable Tumor/Normal Classifier

Project: GeneTransformer · **Date:** 2026-09-13 · **Status:** complete

中文版 / 中文报告: [technical_report_zh.md](#)

Abstract

We set out to predict **cancer type** from bulk gene expression with an interpretable Transformer. A rigorous leave-one-dataset-out (LODO) evaluation revealed that the naive 100 % accuracy was **batch leakage**: each public GEO series is a single cancer type, so platform and class are perfectly confounded, and a random split lets any model memorize the platform. Under LODO the multi-class task collapses to chance (macro-F1 = 0.0), because an unseen cancer type is indistinguishable from an unseen batch. Pivoting to the cross-platform **tumor-vs-normal** task — where the signal is real and generalizes — the Transformer reaches **PR-AUC 0.9963 ± 0.0029 / ROC-AUC 0.9637 ± 0.0192**, on par with a strong RBF-SVM. SHAP and [CLS] -attention attribution reveal a cross-fold consensus gene set dominated by **tumor-stroma / extracellular-matrix remodeling** (collagens, *DCN*, *DPT*, *FBLN2*), with a metabolic component (*PFKFB3*, *LPL*, *PDK4*). Pathway enrichment (Enrichr) confirms this is the **only** signal surviving multiple-testing correction (GO:CC "Collagen-Containing Extracellular Matrix", adj. p = 3.9×10^{-6}). The report documents the confound, the pivot, the domain-shift mechanism that defeats fixed thresholds, and the divergence between the Random-Forest proliferation signature and the Transformer's stromal signature.

1. Introduction & Task

Original task. Classify cancer type (breast BRCA, lung LUAD, colorectal COAD) from bulk microarray expression using a Transformer encoder, and explain each prediction.

The generalization problem. The public data available for these three types are three *different* GEO platforms. Because each platform is also a single cancer type, the class label is aliased with the batch. The central methodological question of the project became: *does the model learn biology, or the batch?* We answer this with the leave-one-dataset-out protocol (§3) and, after it exposed the confound, reframe the task around the signal that genuinely transfers: tumor vs normal (§4).

2. Data

972 bulk **tumor/normal** samples across three independent Affymetrix platforms, after intersecting to a common 14,725-gene matrix:

Platform (GSE)	Tissue	# samples	# normal	# tumor
GSE31210	Lung (LUAD)	246	20	226
GSE39582	Colorectal (COAD)	585	19	566
GSE45827	Breast (BRCA)	141	11	130
Total		972	50	922

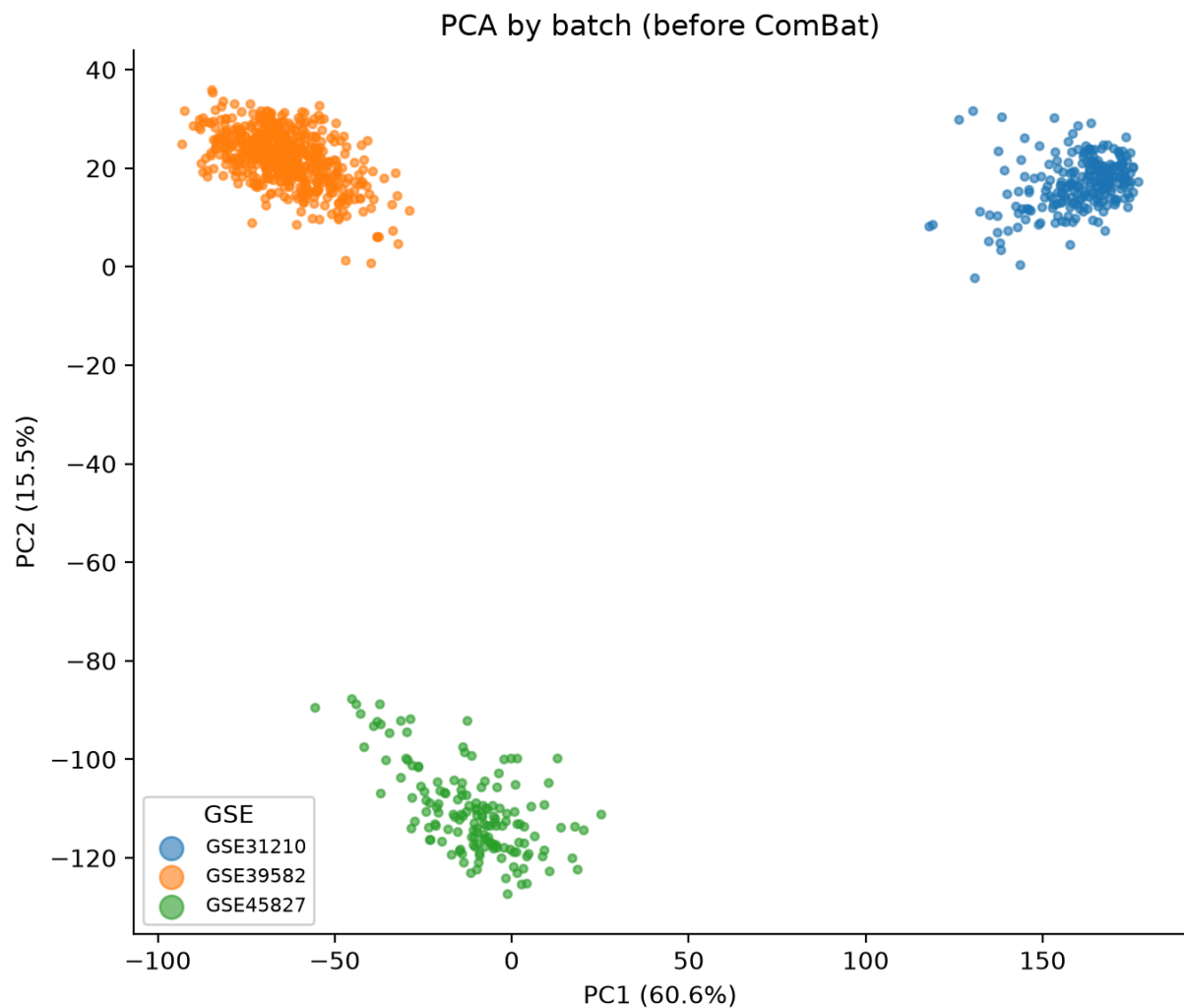
Class imbalance is $\approx 18:1$ (tumor:normal), so we use threshold-free metrics as primary (§5) and up-weight the normal class where a model supports class weights.

Preprocessing. $\log_2(x + 1)$ $\rightarrow$ per-gene z-score $\rightarrow$ **per-fold highly-variable-gene (HVG) selection** (top-5000 by training variance) for the Transformer. Baselines are fit on the same HVG features for a fair comparison.

3. Critical Finding: Batch Leakage

The multi-class task is fatally confounded. We verify it three independent ways:

1. **Batch identity is trivially predictable.** A classifier that predicts *which GSE platform* a sample came from scores **1.0** before batch correction and 0.980 after batch-only ComBat. Cancer identity is also predictable at 1.0 — because *batch == class*.
2. **Random-split control leaks.** With a random train/test split (the setup that produced the original "100 % accuracy"), RF and SVM both reach **macro-F1 = 1.0**.
3. **LODO collapses.** Holding out one entire platform (an unseen cancer type) drives both baselines to **macro-F1 = 0.0** — the model can only assign the held-out samples to a *seen* cancer type, and is always wrong.



Batch and cancer are aliased — PCA before batch correction

Conclusion. The earlier 100 % accuracy was batch effect, not biology. `class == batch`, so ComBat with a cancer covariate is a singular (collinear) design and batch-only ComBat removes the class signal along with the batch. Only LODO exposes the confound, and it shows that *cross-platform multi-class cancer-type classification is not identifiable from these three series*. This is the pivotal negative result that shapes the rest of the project.

4. Pivot: Binary Tumor/Normal

Tumor vs normal is a different signal: it is present in *every* platform and does not depend on which specific cancer type (and therefore which batch) a sample is. We therefore reframe the task as **binary tumor/normal classification under LODO** — the honest, hard generalization benchmark. All subsequent results are on this task.

5. Methods

Models

- **Random Forest** and **RBF-SVM** — strong classical baselines on the HVG features.
- **GeneTransformer** — each gene has a learnable `d=128` embedding scaled by its expression value; a learnable `[CLS]` token is prepended; a 2-layer multi-head self-attention encoder (`n_heads=4` , `d_ff=256` , GELU, post-LN) aggregates context; the `[CLS]` output feeds an MLP head (`→ 64 → 2`).

Evaluation

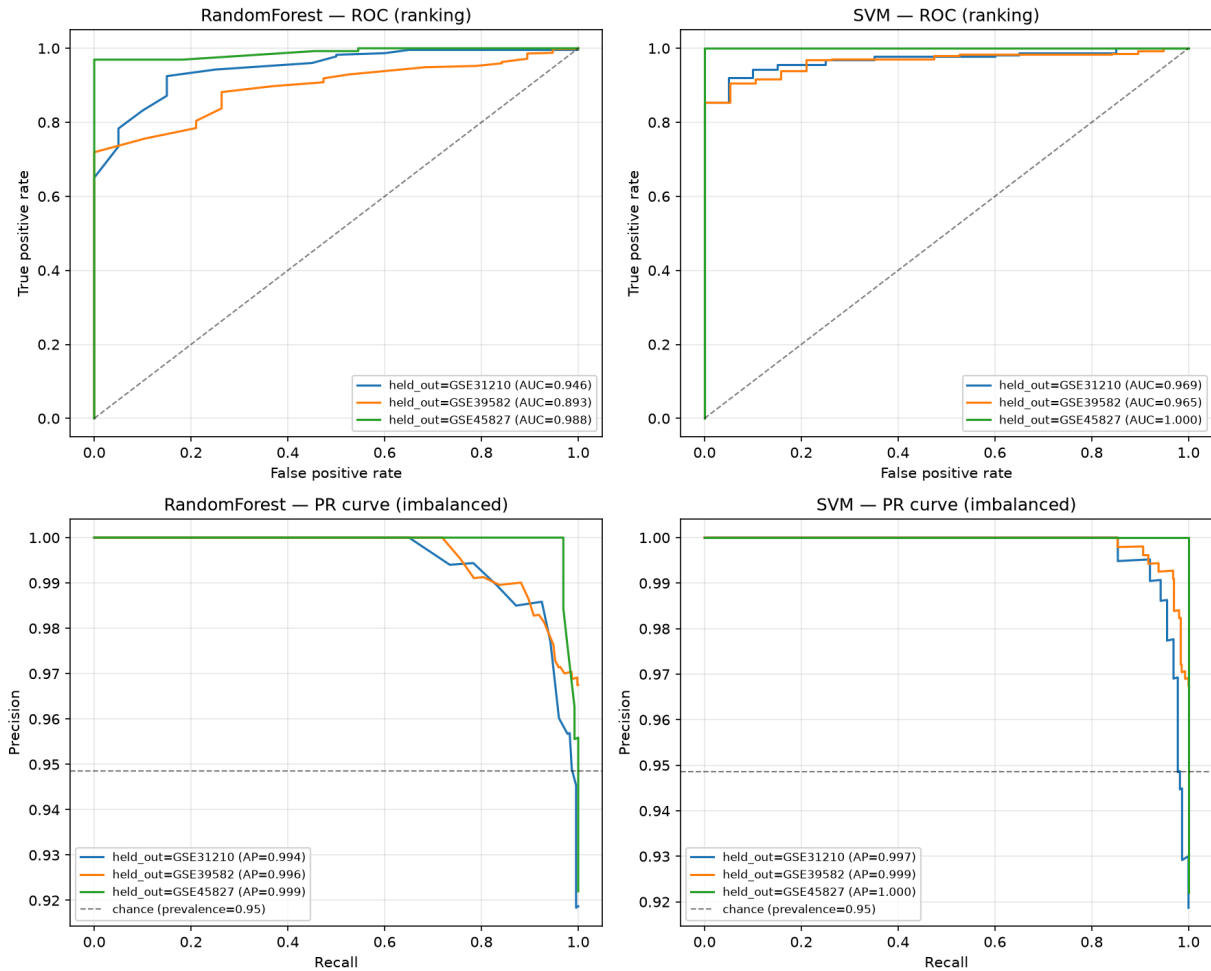
- **Primary metrics are threshold-free: PR-AUC and ROC-AUC.** Accuracy / F1 / balanced-accuracy are abandoned for this task: under $\approx 18:1$ imbalance and domain shift they collapse to a misleading "predict-tumor-always" 0.5 even when the ranking signal is real.
- **Hard labels (secondary)** use **quantile alignment**: each held-out fold's score is thresholded at the *training* fold's normal-prevalence quantile — label-free on the test set, and the only threshold that survives the cross-platform score shift (§6).
- **LODO** holds out one whole platform per fold; every test fold is unseen in training.

6. Results — Binary Classification

Model (LODO)	PR-AUC (mean $\pm$ std)	ROC-AUC (mean $\pm$ std)
Random Forest	0.9961 $\pm$ 0.0025	0.9486 $\pm$ 0.0250
SVM (RBF)	0.9986 $\pm$ 0.0012	0.9776 $\pm$ 0.0154
Gene Transformer	0.9963 $\pm$ 0.0029	0.9637 $\pm$ 0.0192

The Transformer matches the SVM on PR-AUC and exceeds the RF on ROC-AUC, while remaining end-to-end differentiable — the property that makes §7's attribution possible.

Tumor vs normal — LODO per-fold ROC & PR

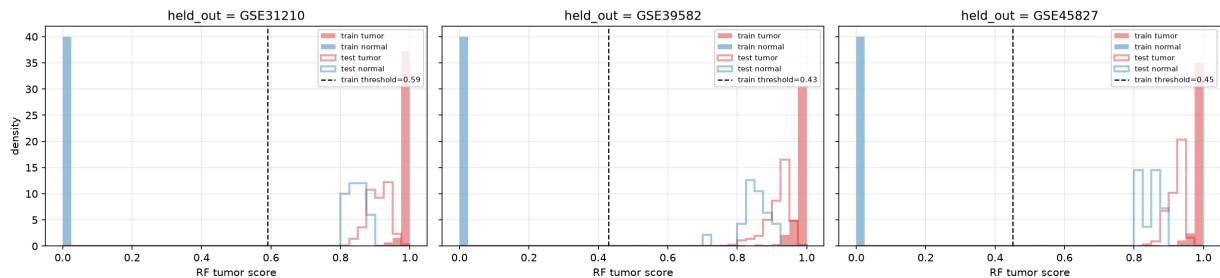


ROC / PR curves (LODO)

Domain shift defeats fixed thresholds

The tumor/normal *ranking* transfers across platforms, but the *score scale* does not. On the training fold the model saturates (tumor ≈ 1.0 , normal ≈ 0.0); on the held-out fold the normal scores shift up toward the tumor band (test normal median ≈ 0.85), so any threshold learned on the training fold misclassifies most held-out normals.

Score-shift diagnosis: train-fit threshold does not transfer cross-platform



Score shift: test normal scores drift toward tumor across platforms

Quantile alignment fixes this: normal recall rises from ≈ 0 under a fixed threshold to 0.54 (RF) / 0.59 (SVM). Crucially, `class_weight` (even 18:1) $\approx$ balanced does **not** fix it — the shift is a

score-scale offset induced by cross-platform technical variation, not a class-prior problem.

7. Interpretability — SHAP & Attention

Two complementary, inference-only views of *why* the Transformer decides (no retraining; run on CPU from local checkpoints):

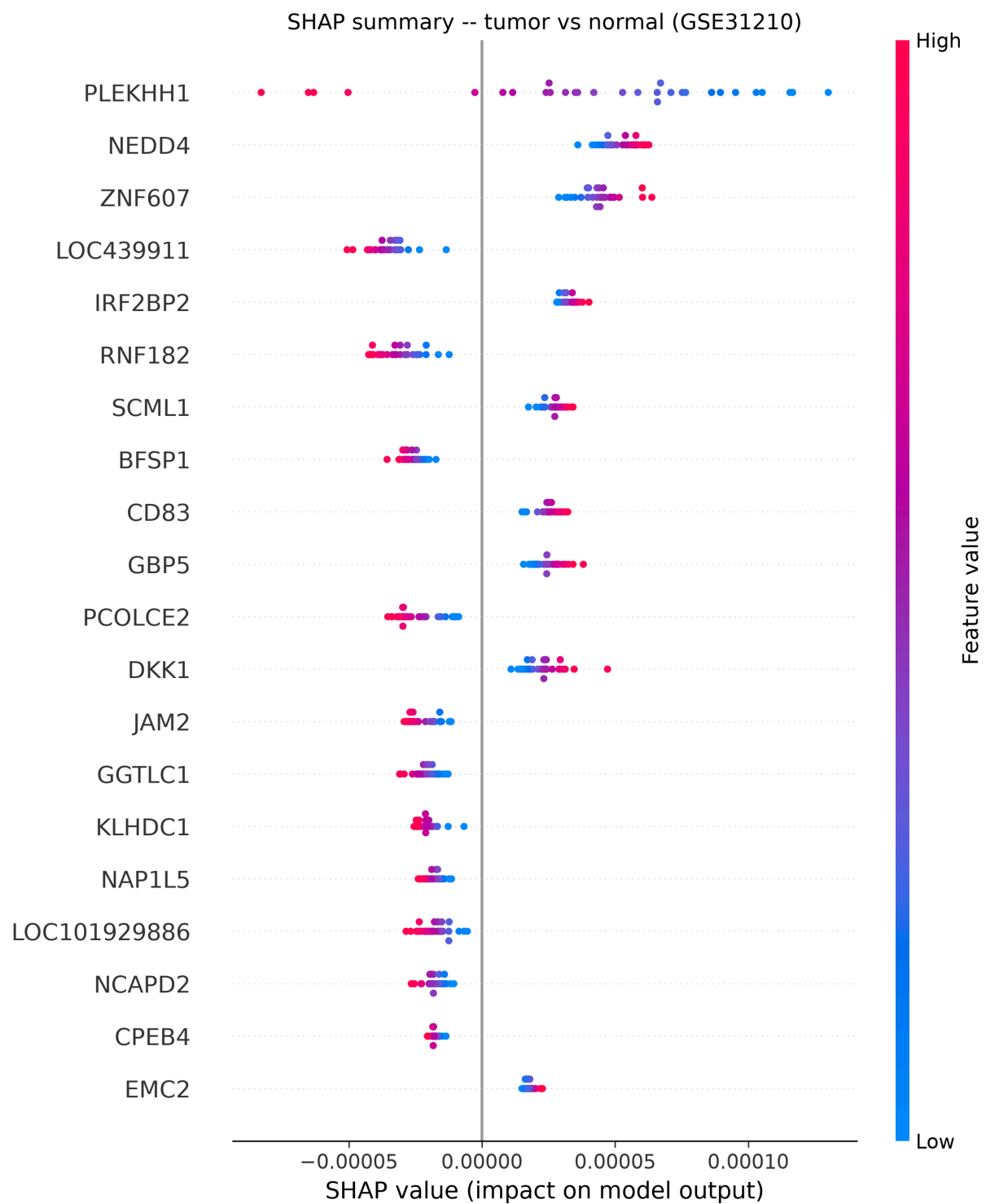
- **SHAP (Deep SHAP / Gradient × Input).** For this architecture `shap.DeepExplainer` reduces to `SHAP[s,i] = (x[s,i] - mean(bg)[i]) · mean(d logit / d x_i)`, background = stratified train mean, target = `logit[tumor] - logit[normal]`. Computed in `batch=1` chunks because full self-attention over 5001 tokens is $O(n^2)$ and ~5 GB RAM would OOM.
- **Attention ([CLS] → gene).** The [CLS] token's self-attention row is recomputed per layer/head from the layer's Q/K projection weights (`_cls_attention`), materializing only the CLS row ($O(L)$) — exact to 1e-9 vs `need_weights=True` — averaged over layers, heads, and test samples.

Cross-fold aggregation is rank-based with an absent-fold penalty: each gene's per-fold importance becomes a rank (1 = best); a gene absent from a fold's HVG set gets the worst rank (`n_genes+1`); the two methods' mean ranks are summed. This rewards genes that are *consistently* important across platforms and is robust to the heavy-tailed raw scores.

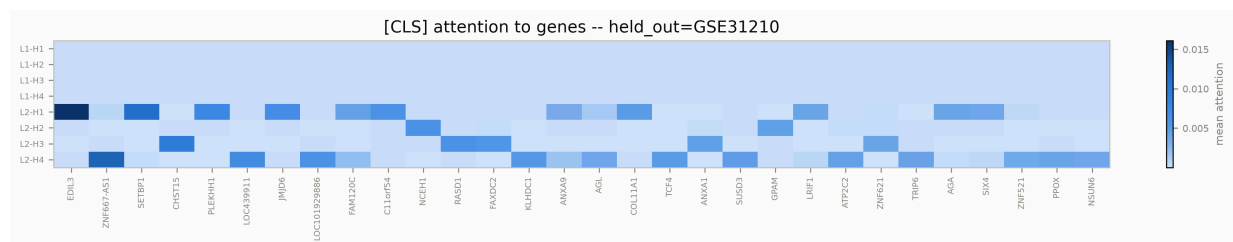
Consensus genes (top-20, rank-based, `n_folds=3`)

ZNF521, PFKFB3, COL4A5, LPL, AFAP1-AS1, CDO1, DPT, LCN2, COL11A1, ABCA8, ACADSB, CLTC, C2CD4A, EBF1, FLJ13744, LEPR, AR, EGR3, CA4, DCN

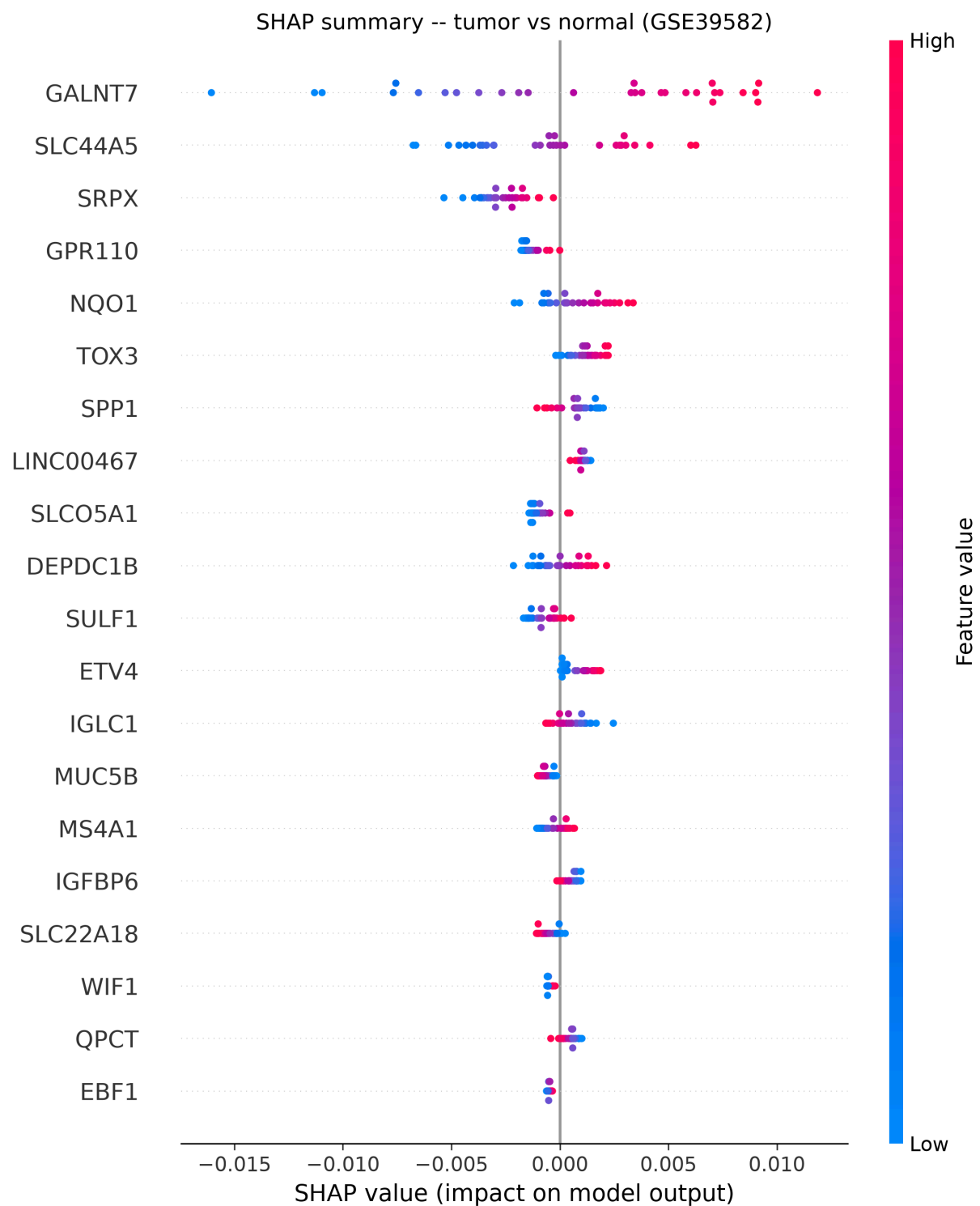
These cluster into **extracellular-matrix / stroma remodeling** (*COL11A1, COL4A5, COL10A1, DCN, DPT, FBLN2, EFEMP1, CCDC80, ABI3BP, OGN, CLEC3B, EDIL3*) and **metabolism / signaling** (*PFKFB3* glycolysis, *CDO1* methylation marker, *LCN2, LPL, PDK4, ACADSB, GPAM*). Per-fold top genes are markedly platform-specific (GSE31210 → PLEKHH1/NEDD4; GSE39582 → GALNT7/SLC44A5; GSE45827 → CST1/GPR110/SCARA5), mirroring the domain shift — the consensus is what survives all three platforms.



SHAP summary — GSE31210 (clearest fold #1)



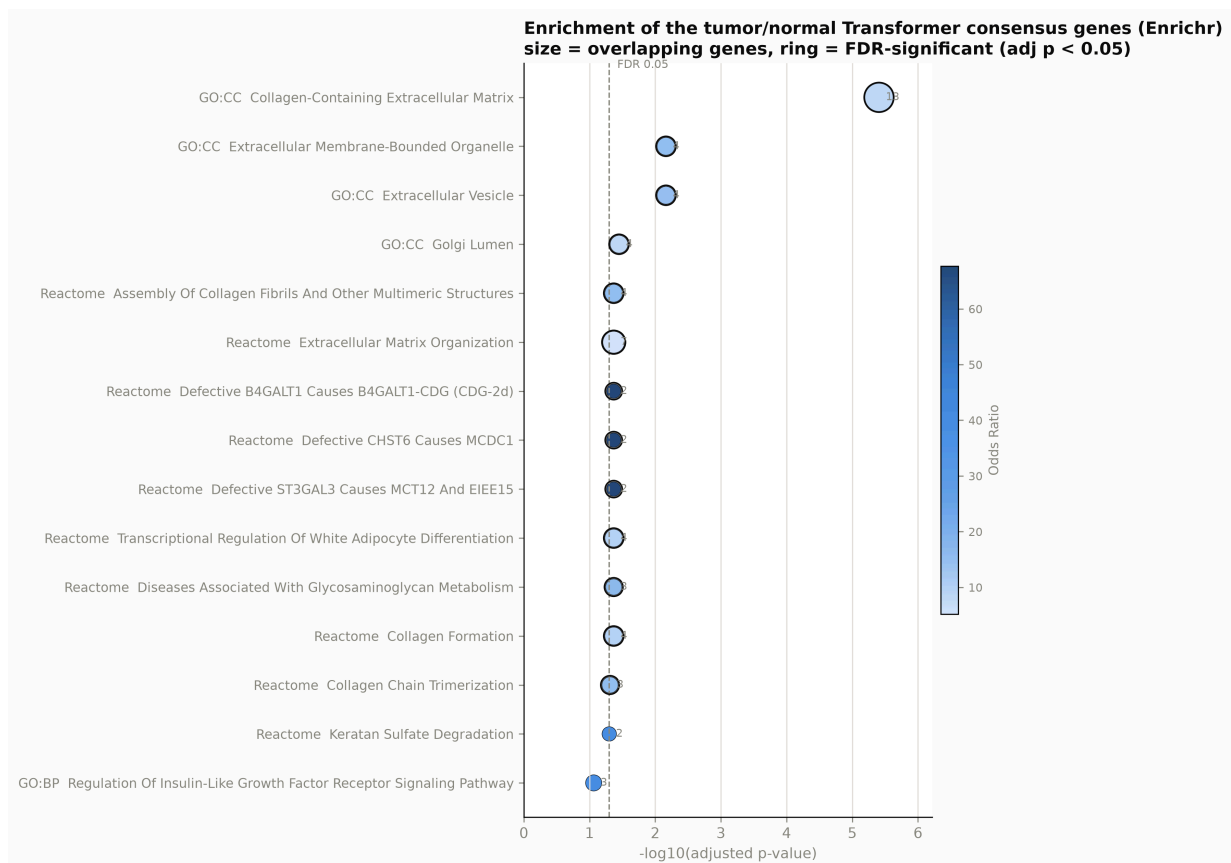
CLS attention — GSE31210 (clearest fold #2)



SHAP summary — GSE39582 (clearest fold #3: different genes dominate)

8. Pathway Enrichment (GO / KEGG / Reactome)

Top-100 consensus genes → Enrichr ([KEGG_2021_Human](#) , [GO_BP/MF/CC_2023](#) , [Reactome_2022](#)).



Enrichment bubble plot

The one FDR-significant signal is extracellular-matrix / collagen organization:

Library	Term	adj. p	overlap
GO:CC	Collagen-Containing Extracellular Matrix	3.9×10^{-6}	13
GO:CC	Extracellular Vesicle / Membrane-Bounded Organelle	0.0069	4
Reactome	Assembly of Collagen Fibrils & Other Multimeric Structures	0.043	4
Reactome	Extracellular Matrix Organization	0.043	7
Reactome	Collagen Formation / Chain Trimerization	0.043–0.049	3–4

KEGG, GO Biological Process, and GO Molecular Function produce **no** term surviving multiple-testing correction — their strongest nominal hits (GO:BP "ECM Organization" raw $p = 2.6 \times 10^{-4}$, adj. $p = 0.088$; KEGG "AMPK signaling" raw $p = 0.022$, adj. $p = 0.48$; KEGG "Pathways in cancer" adj. $p = 0.48$) are trending only. We report this **honestly**: the consensus gene set is a **tumor-stroma / ECM-remodeling signature** with a secondary metabolic component — not a classical oncogene/tumor-suppressor pathway.

9. Discussion

1. **The Transformer sees the tumor microenvironment.** Its most consistent genes are stromal/ECM — collagens, decorin, dermatopontin, fibulins — a hallmark of tumor desmoplasia, rather than a proliferation program.
2. **Low overlap with the RF proliferation signature.** The RF's top genes are proliferation markers (*TPX2*, *ECT2*, *UBE2C*, *UBE2T*, *PPAT*, *BOP1*); in the Transformer they rank only ~1000–3000 (SHAP) / ~400–900 (attention), and several are absent from multiple folds (`n_folds < 3`). This is expected: **(a)** HVG selects genes by *variance*, and low-variance proliferation genes are frequently dropped from the 5000-gene Transformer input while the RF sees all ~15 k genes; **(b)** the two models have different inductive biases. The Transformer's HVG input predisposes it to high-variance stromal signals.
3. **Domain shift, not class prior, governs hard labels.** Under LODO the score scale shifts between platforms; only quantile alignment yields usable hard labels. This is a property of the data/feature-selection, not of the architecture.
4. **The pivotal result is the negative one.** Cross-platform multi-class cancer-type classification is not identifiable from these series (`batch == class`); the binary tumor/normal signal is. Any future work should source data where `batch != class` (e.g. TCGA — one platform, multi-cancer + matched normal) before revisiting multi-class.

Limitations

- Interpretability runs on 30 stratified test samples per fold (memory-bounded), so per-fold rankings are indicative, not exhaustive.
- Enrichment foreground is the top-100 consensus by *rank*, not a differential-expression test; background is Enrichr's default (all human genes). LncRNA/loci (AFAP1-AS1, FLJ13744, LOC285628, ...) do not map to pathways and are silently excluded.
- The Transformer's AUC trails the SVM slightly; we did not tune it aggressively, since the project's goal was interpretability, not beating the SVM.

Reproducibility

All hyperparameters and paths live in `config.yaml`; the seed is fixed (42). Commands:

```
python -m src.baseline, python -m src.train, python -m src.explain_tn,  
python -m src.enrichment. Checkpoints, preprocessed matrices, and raw data are gitignored  
(reproduced, not committed).
```